From Failure Discovery to Action Evaluation: A Controlled StressDream Research Investigation

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Independent technical investigation; unpublished · 22 October 2026

1. Research question and motivation

When does failure-oriented world-model steering provide reliable action-level evaluation? Does better possible-failure discovery improve true-dynamics action ordering and regret at fixed actions and total compute? The completed oracle, design checks and compatibility/readout pilot do *not* establish that relationship or a new planning algorithm.

2. Relation to StressDream

Paper-reported evidence. StressDream optimizes diffusion noise to expose high-impact, plausible futures in frozen action-conditioned world models [1]. It already includes ground-truth Dubins failure checking, plausibility regularization/ablations, candidate-action examples and policy improvement. The present question concerns measurement within that established agenda, not an absence of decision evaluation.

Source observations. Revision `b16fa067d3d9` [2] raises flip-probability, deterministic-demo, momentum, regularizer and score/noise/sample-alignment questions. Their impact on learned-model performance is unmeasured. The steered ranking comparison remains proposed (Figure 1).

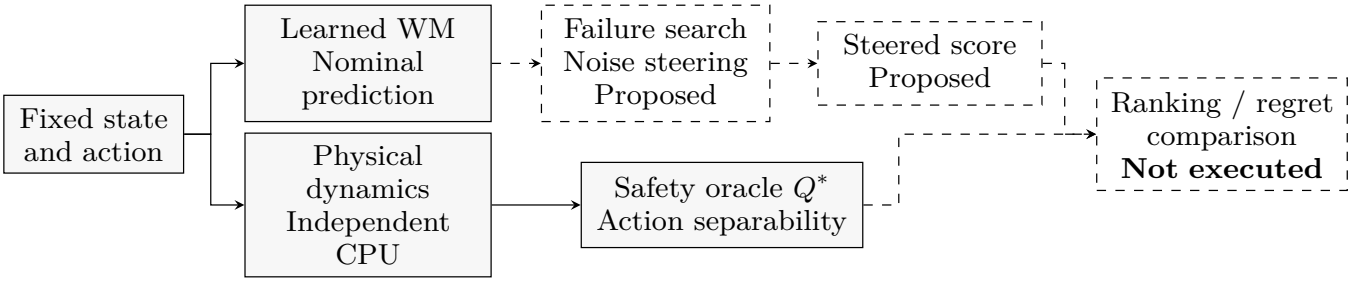


Figure 1: Solid: completed oracle or nominal interface work. Dashed: proposed steering and action-ranking comparison. The schematic separates experimental contracts; it does not depict network architecture.

3. Independent finite-horizon dynamics oracle

The float64 reference implements the released discrete stochastic Dubins dynamics:

$$\mathbf{p}_{t+1} = \mathbf{p}_t + v\Delta t(\cos\theta_t, \sin\theta_t), \qquad \theta_{t+1} = \theta_t + \Delta t\, \delta_t a_t.$$

Here $v = 1\,\mathrm{m/s}$, $\Delta t = .05\,\mathrm{s}$, $|a_t| \le 1.25\,\mathrm{rad/s}$ and independent $\delta_t \in \{-1, +1\}$. Position uses the *old* heading. At $H = 12$, all $2^{12} = 4{,}096$ sign strings are enumerated per query, including safety at $t = 0$:

$$C(s) = x^2 + y^2 - .25^2, \qquad Q^*(s,a) = \min_{\delta} \min_{0\le t\le 12} C(s_t), \qquad R(s,\hat{a}) = \max_{a} Q^*(s,a) - Q^*(s,\hat{a}).$$

Larger Q^* is safer; failure is $C \le 0$; the released danger head uses $m = -C$. Sixteen unit tests passed, including scalar checks, signs, update order and witnesses [3]. "Exact" means exhaustive *discrete* support in floating point, not certified real arithmetic or continuous-time safety.

4. Gate 0: separation with weak decision diversity

All 16 states and five actions remain frozen. At $0.005\,\mathrm{m}^2$, **12/16 states** are informative and **106/160 pairs** distinguishable. Yet constant straight-line A0 is practically optimal: maximum robust regret is approximately $0.000176\,\mathrm{m}^2$. The pool supports ranking diagnostics but weakly tests state-dependent choices. This is an oracle design finding, not evidence about StressDream reliability.

5. Gate 0B: a separately declared task extension

A separate design pool fixed an unfiltered 64-context held-out test with multi-side states and signed controls. Gate 0B **changes the objective**: expected squared-distance progress toward $(1, 0)$ at flip probability .2, subject to support safety $Q^* > 0$ [3]. At $0.005\,\mathrm{m}^2$, 43 contexts are informative; 41 have meaningfully unique optima, spanning all five IDs. Best-constant mean *task* regret is $0.103087\,\mathrm{m}^2$ (unsafe score: $-4\,\mathrm{m}^2$); all 21 no-safe-candidate contexts remain included. This extension is not StressDream's original objective; its regret scale is not directly comparable with Gate 0.

6. Gate 1A: compatibility and independent readout

Both official checkpoints passed size/hash checks, restricted CPU loading and architecture compatibility; prediction-head weights were present [4]. The nominal CPU pilot completed 43 future predictions, 32 paired VAE reconstructions and 215 denoiser forwards in approximately 38.0 model-process seconds, with 0.318 GiB sampled aggregate peak RAM. These are not steering/full-benchmark costs.

A pixel-only reader, frozen after separate tuning, parsed 16/16 validation images in each paired domain. Position p95 errors were 13.92 mm clean and 13.97 mm reconstructed. The reader is not yet accurate enough for the proposed ranking comparison (Figure 2). For position error bounded by e, $|\widehat{C} - C| \le 2\|\widehat{\mathbf{p}}\|e + e^2$.

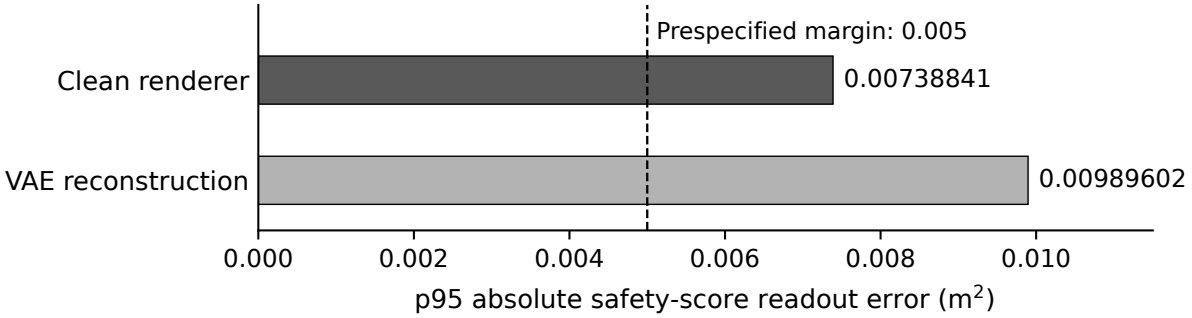


Figure 2: Readout p95 on 16 paired validation states: clean 0.00738841, VAE $0.00989602\,\mathrm{m}^2$. Dashed: prespecified ranking margin $0.005\,\mathrm{m}^2$; per-action readout uncertainty target: $0.00125\,\mathrm{m}^2$. No population interval or world-model failure claim.

7. Evidence boundaries

Compatibility **passes**; Gate 0 ranking remains **conditional**; Gate 0B expected progress is **no-go at present**. Both nominal $H = 12$ paths missed a coarse consistency tolerance, with model/readout errors confounded. Training flip probability is undocumented (paper .2, generator .4): common support does not imply equal nominal expectations, and steered sample fractions cannot estimate them.

No steering, full ranking or policy-improvement experiment ran. A final export error lost some tensor/range and future-head logs; retained evidence was recovered without inference replay. No ranking degradation, world-model failure or new algorithmic contribution is established.

8. Proposed next experimental step — not executed

First validate geometry below the frozen score margins. A separately approved nominal/Best-of-N/steering comparison would fix states/actions, align noise/latent/score records and match *total* per-action cost (backward passes, regularization, decoding, rescoring). Report supported-failure recall separately from tie-aware ranking/regret. Expected progress also needs probability calibration.

9. References and reproducibility

[1] J. Seo et al. *StressDream: Steering Video World Models for Robust Policy Evaluation and Improvement.* CoRL 2026; arXiv:2606.00267v2.

[2] CMU Intent Lab. StressDream source, revision `b16fa067d3d949ef09ad48a984c62ea8ecbbfe6d`.

[3] Z. Wu. *Dubins Oracle and Decision-Diversity Audits* (Gate 0/0B), 2026. Unpublished audit records.

[4] Z. Wu. *Checkpoint/Readout Compatibility Pilot* (Gate 1A), 2026. Unpublished audit records.

Public reproducibility: `https://github.com/ZhijingWu/StressDream-Decision-Reliability-Audit`. Frozen configurations, commands, numerical evidence and SHA-256 mappings are public; selected reports are sanitized copies. This note adds no experiments or complete paper reproduction.